In the following example, the scores are calculated using the singular value decomposition; so the Figures 12 and 13 are equal among them but rotated compared with Figure 6. This has been explained at *http://stackoverflow.com/questions/17998228*, where, to quote user Hong Ooi: "The signs of the Eigenvectors are arbitrary. You can flip them without changing the meaning of the result, only their direction matters."

```
import mdp
pca=mdp.nodes.PCANode(svd=True)
scores=pca.execute(array(dataS))
```

The MDP package is more complex than described here. Many other things can be done with it. It's also well documented; for example, the tutorial is more than 200 pages long.

The examples presented here are also typical applications for another, very widely used, free and open source software, R. An interesting comparison of Python and R for data analysis was published some time ago (Reference 7). I can't make a choice, because I like them both. Currently, I use Python almost exclusively, but in the past, R was my preferred language. It's useful to develop the same script for both and then compare the results. END

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[4] *Zito, Wilbert, Wiskott, Berkes, Modular toolkit for Data Processing (MDP): a Python data processing framework, Frontiers in Neuroinformatics, 2009, 2, 8.*
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@pyimport is used to import the sub-modules of Python.

Julia packages

Another interesting feature of Julia is the extensibility similar to what we have in languages like Python. As of July 3, 2016, the official repository has 1035 packages. The evolution of the package ecosystem *(http://pkg.julialang.org/pulse.html)* is illustrated in Figure 6.

Primarily, packages enable us to extend the functionality of the core Julia engine. The areas covered by the packages are really diverse in nature. There are packages available for mathematical operations, machine learning, GUI, etc.

To add a package, enter the following command at the Julia prompt:

```
Pkg.add(" package name")
Pkg.add("PyPlot")
```

To use the package, the *using* keyword is employed:

```
using PyPlot
```

Plotting with Julia

Julia facilitates plotting as well. A sample code is shown below:

```
using PyPlot
x = linspace(0,2*pi,1000); y = sin(3*x + 4*cos(2*x));
plot(x, y, color="red", linewidth=2.0, linestyle="--")
title("A sinusoidally modulated sinusoid")
```

The output plot of the above code is shown in Figure 7. (The successful execution of this code requires Python and MatPlotLib availability in the system.)

Julia is certainly a promising programming language, and its use is on the rise among scholars and data scientists. In the years to come Julia will evolve further; so try and get familiar with its ecosystem. The Julia home page has plenty of resources in the form of PDFs, video lectures, live notebooks, books, blogs, etc, which makes the process of getting insights into the Julia development process simple, efficient and enjoyable. Computing with Julia can really be awesome. Give it a try. END

References

[1] *http://julialang.org/downloads/platform.html*
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